Assignment
CSL251

Q.1 Design an **8:1 Multiplexer** with:

- **Inputs:** `I0, I1, I2, I3, I4, I5, I6, I7` (each 1-bit)
- **Select lines:** `S2, S1, S0` (3-bit)
- **Output:** `Y`

Verilog Code ,Test Bench and output screen short.

Q.2 Design a **2-bit ALU** with:

- **Inputs:**
  - `A[1:0]` (2-bit)
  - `B[1:0]` (2-bit)
- **Select lines:** `S1, S0`
- **Outputs:**
  - `Y[1:0]` (Result)
  - `Cout` (Carry/Borrow)

Verilog Code ,Test Bench and output screen short.